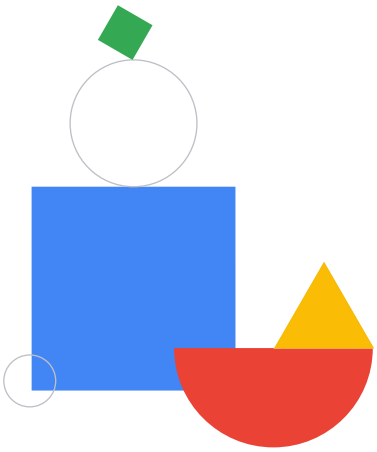
Google Cloud

Introduction to Looker as a Data Platform



Welcome to the module on Introduction to Looker as a Data Platform.

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Agenda

01	Demo: Looker UI Overview
02	Looker Platform Architecture
03	Paradigm Shift: Modeling Language vs. Hardcoded SQL
04	Core Analytical Concepts **Lab:** Getting Started with Analyzing and Visualizing Data in Looker

Google Cloud

In this module, we'll introduce the key features of Looker that empower organizations to leverage their data for real-time business analytics.

We'll start with a demonstration of Looker's user interface, and then go over the Looker platform architecture.

Next, we'll review the paradigm shift in Looker from hardcoding SQL toward the use of LookML (Looker modeling language).

Last, we'll explore the core analytical concepts that will enable you to start analyzing and visualizing data in Looker.

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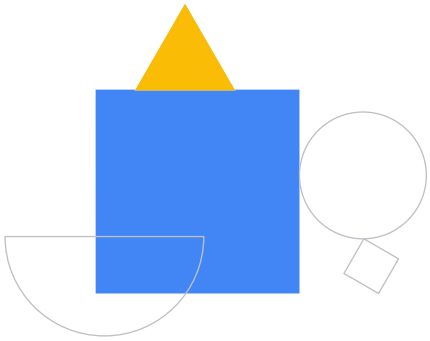
Demo

Looker UI Overview



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Demo video:
https://storage.googleapis.com/cloud-training/datawarehousing/v1.2.1/en/Demos/PTI-PWDW-A_M6_I2_demo_looker_ui_overview_v3.mp4

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In this section, we'll explore the Looker platform architecture to understand how it enables self-serve data exploration, analysis, and visualization within an organization.

Let's begin with a few key characteristics of Looker as a data platform.

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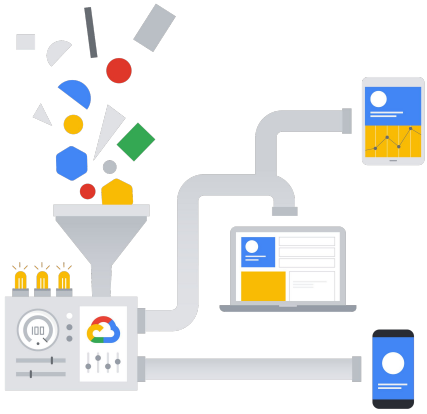
Business intelligence (BI) is a class of software that aims to turn a company's data into valuable insights



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Business intelligence (BI) is a class of software that aims to turn a company's data into valuable insights for both internal and external audiences.

Looker is a BI software and big data analytics platform that helps you explore, analyze and share real-time business analytics easily.

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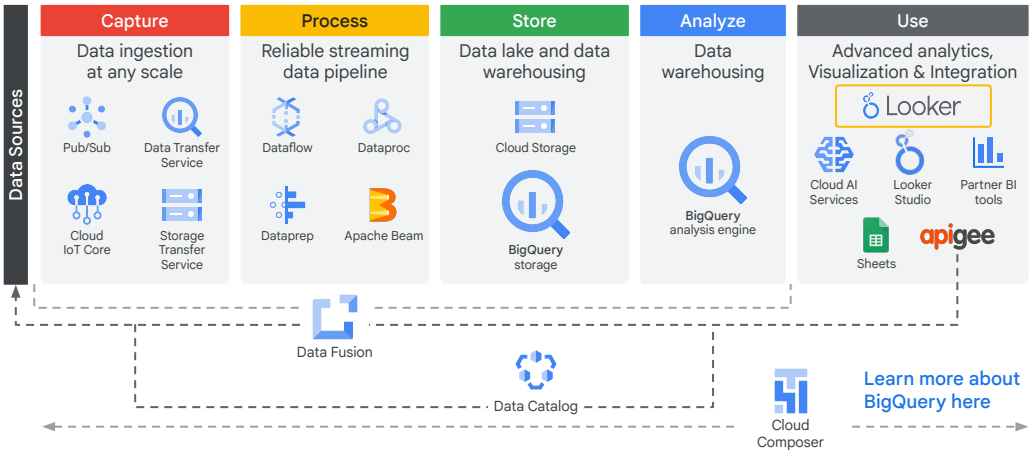
Looker is the Google Cloud enterprise platform for data experiences



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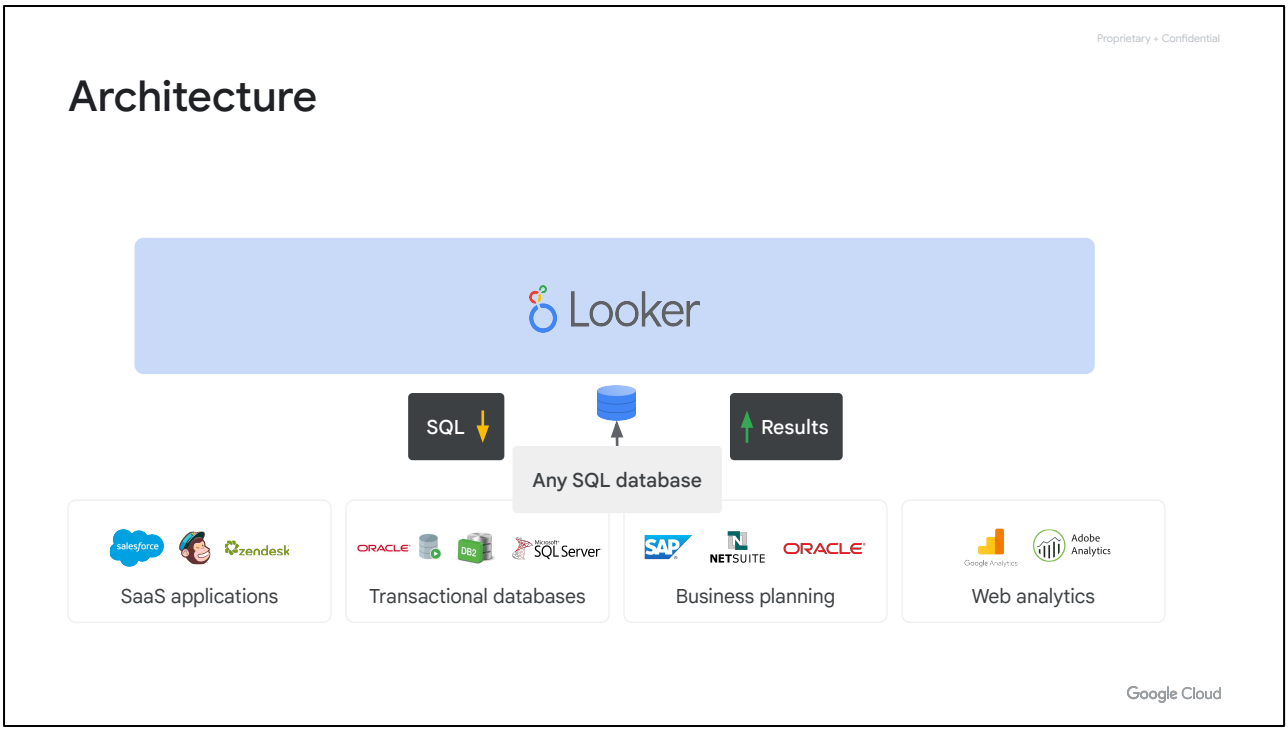
In addition, Looker is Google Cloud's enterprise platform for data experiences — from internal business intelligence to externally embedded analytics to custom data applications.

Looker's unique architecture provides a rich development framework that is built to support enterprise-grade workflows and use cases.

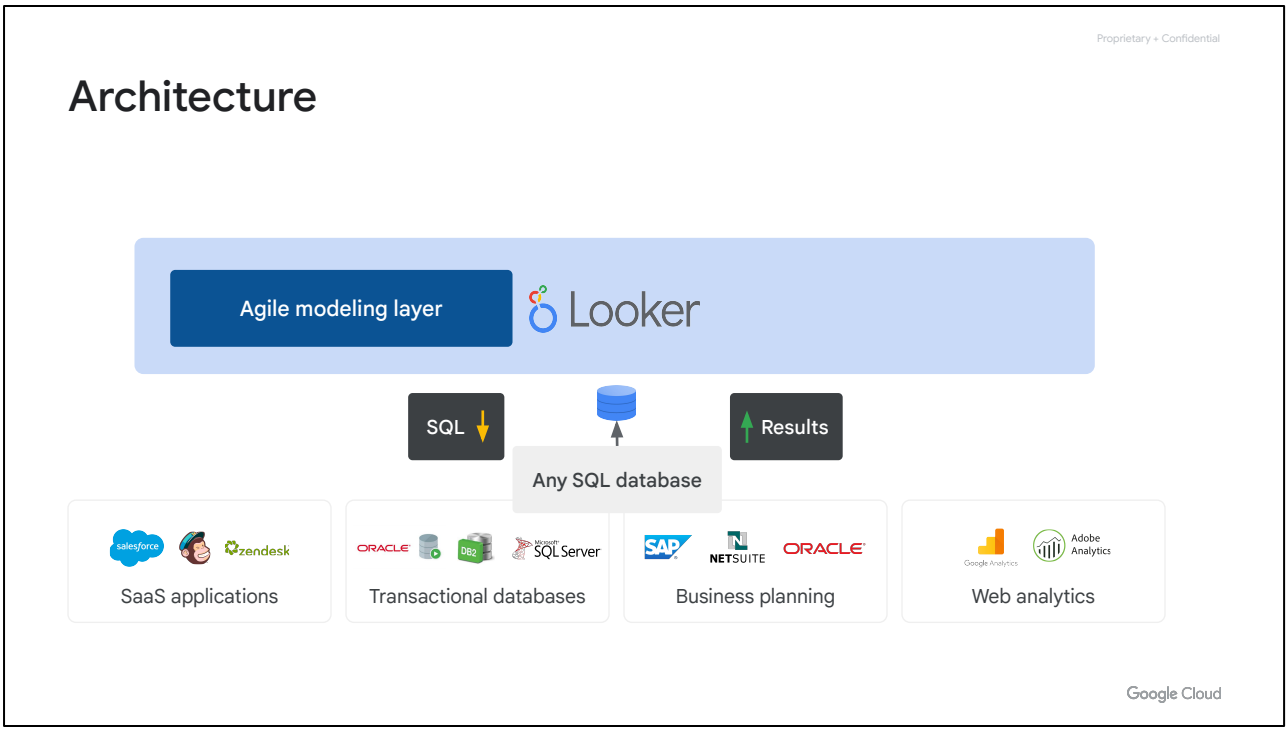
With Looker, organizations can drive better outcomes, enable smarter data experiences, and find true power in their data. Organizations can move beyond traditional, stale BI and infuse fresh data into every workflow.



Looker is a browser-based platform that connects directly to SQL databases. For example, you can link Looker to:

- Software as a Service (or SaaS) apps such as Salesforce, Mailchimp, and Zendesk.
- Heavy read-write operations in transactional databases such as Oracle, IBM Db2, and Microsoft SQL Server.
- Business planning tools such as SAP, NetSuite, and Oracle.
- And web analytics products such as Google Analytics or Adobe Analytics.

These are just a few examples. Looker is multi-cloud and supports over 65 database dialects.



One major benefit of Looker is its agile modeling layer, which can save data teams and business analysts time that would otherwise be spent manually writing and editing SQL queries.

Looker's agile modeling layer allows developers to define, through LookML, how the database is structured and how the tables and columns relate to each other. As business users explore and analyze data, Looker uses these LookML models to automatically construct SQL **SELECT** queries to send to the database and return the appropriate results.

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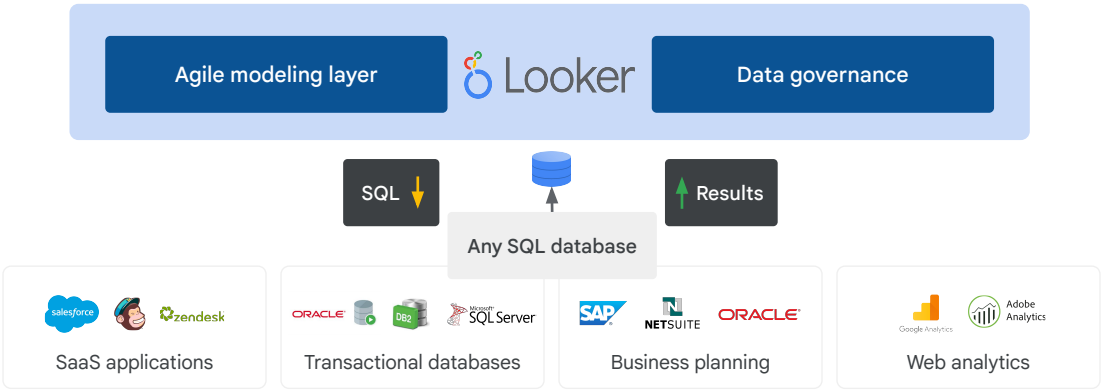
Architecture



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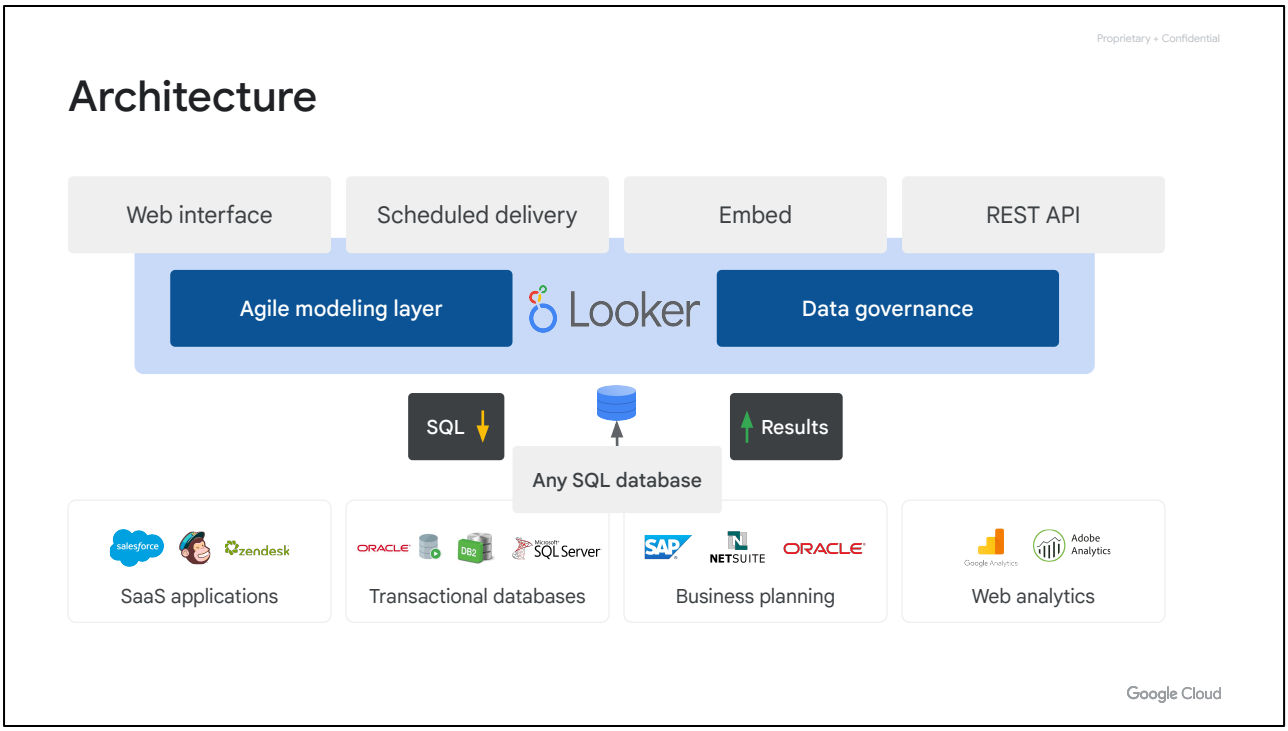
Another benefit of Looker is *data governance*, which means that you can define a single source of truth for data that everyone in the organization can understand and trust. You can also enforce various types of data security and governance within the Looker platform and using LookML.



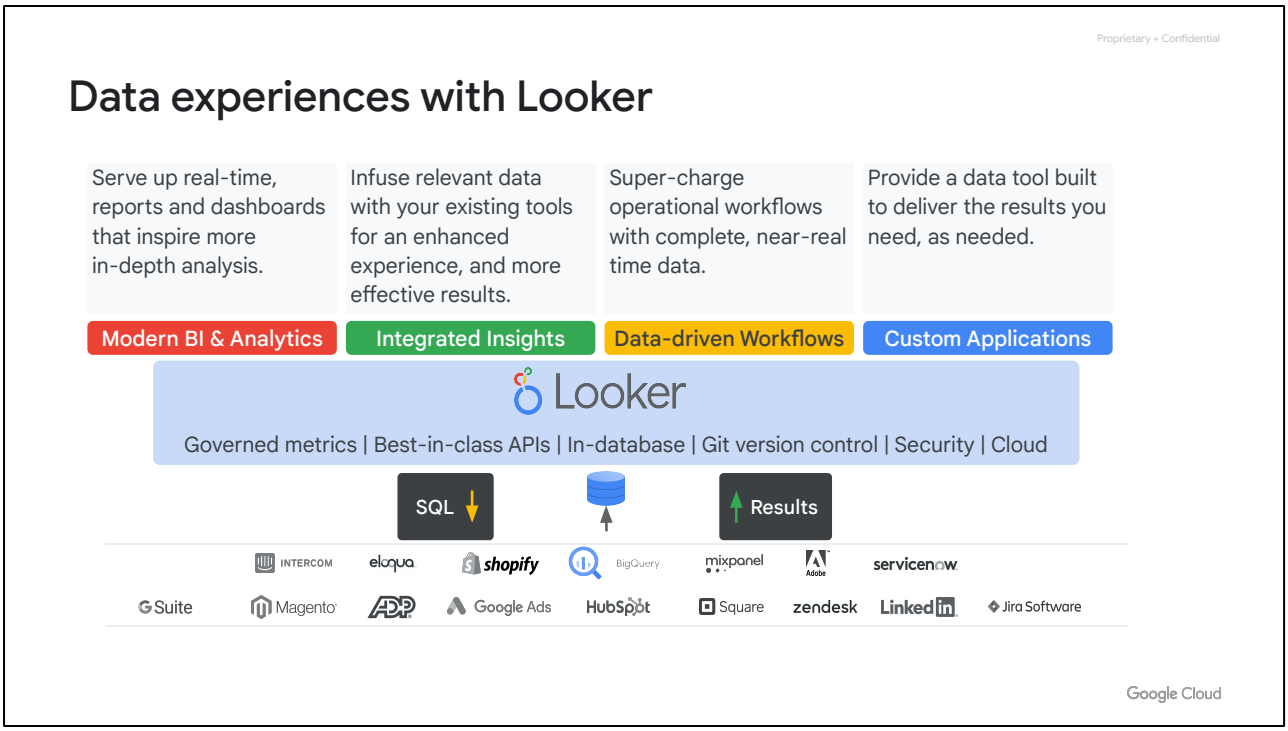
To help organizations disseminate data, Looker can surface and expose query results in several ways.

The first is through the *web interface*, in the form of Explores (which are report-builder interfaces), Looks (which are standalone reports or visualizations), and dashboards (which contain multiple visualizations).

Another way is through *scheduled data deliveries*, such as sending Looks and dashboards to specific email addresses or Cloud Storage buckets on a one-time or recurring basis.

Third, Explores, dashboards, and Looks can be *embedded* in other websites and applications.

And last, Looker provides a REST API that allows you to retrieve, analyze, and transform data and metadata directly from the Looker platform.



Looker powers a number of data-driven experiences, beginning with modern BI. Looker's world-class BI experience puts trusted, actionable insights into the hands of decision-makers and helps cultivate a data culture throughout the organization.

With modern BI, companies can serve up real-time, relevant reports and dashboards that act as a starting points for more in-depth analysis. Easy access to trustworthy data enables people to make better, data-driven decisions every day.

In addition, with fresh, accurate data on hand, it's easy to optimize workflows with data. With complete, near-real time data, these data-driven workflows will save time and money by putting your data to work in every part of your business. You can also integrate Looker with your existing tools, or build custom applications to deliver the most appropriate results to your different audiences.

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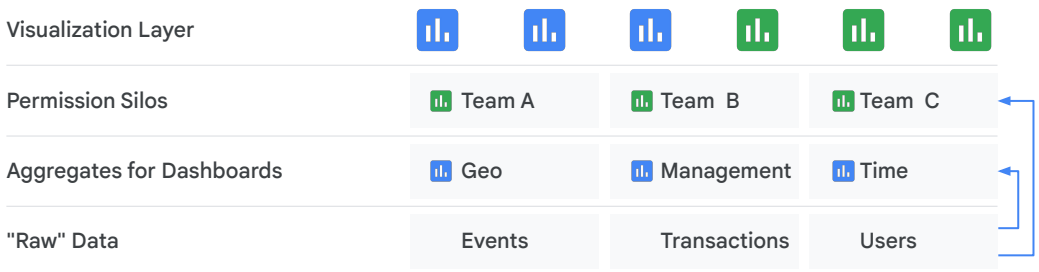
Typical workflow for BI and visualization tools



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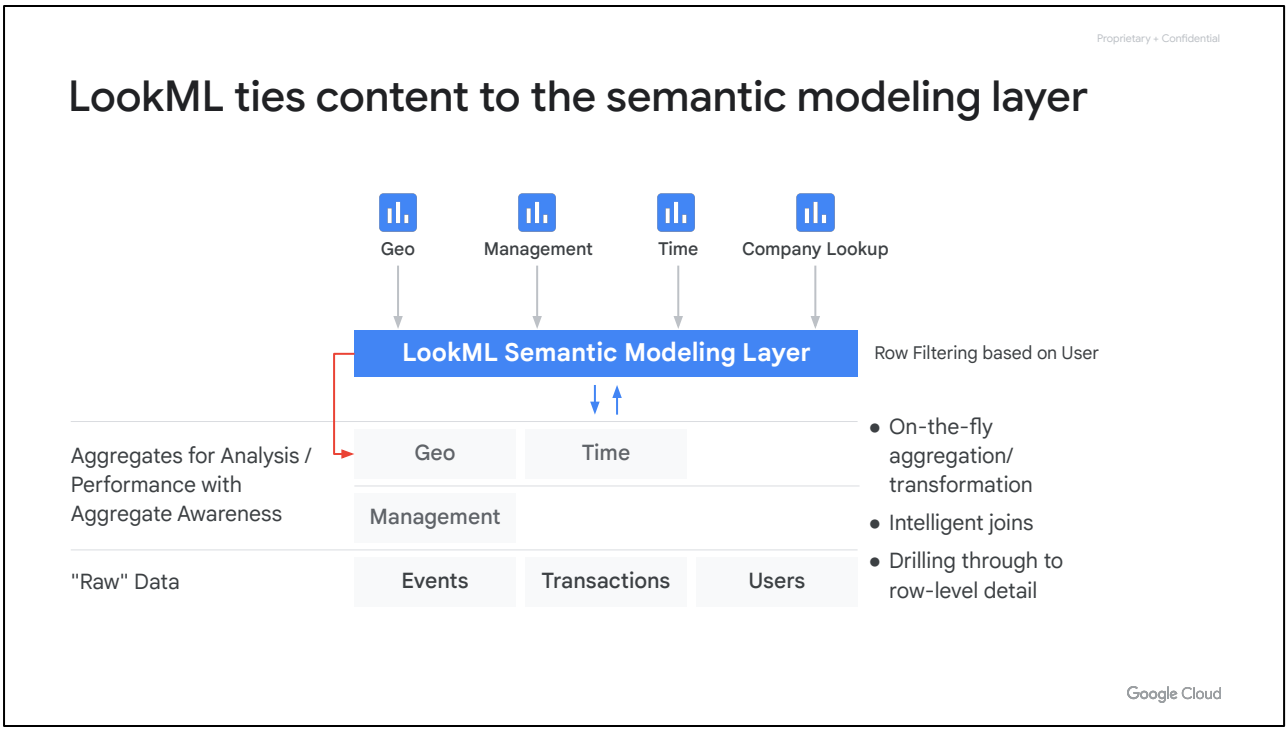
To fully understand the benefits of Looker's agile modeling layer, it is helpful to review how it compares to the typical workflow for traditional BI tools.

A common workflow for traditional BI tools begins with the design team, who will talk to the business stakeholders, come up with dashboard wireframes, and go to the database administrators (or DBAs) with data requirements needed to build the desired dashboards.

The DBA then creates the metrics or aggregates needed for those dashboards—in other words, they are aggregating data just for that project—thereby possibly creating permission and access silos.

The result of this workflow is that you can end up building a dashboard off a very wide table, or off some subsets of data that you've prepared just for that dashboard. As requirements and end users change, you would need to loop through this cycle and identify new database tasks for the DBAs.



With Looker, you start in the same place - the database. However, rather than going to the DBA and design team, LookML developers add a modeling layer on top of the raw data —and then tie the final content to this semantic modeling layer. Thus, end user content isn't tied straight to the database, but rather to the LookML model files.

This gives not only database admins or data engineers, but also data and business analysts, the flexibility to create content themselves using the data options provided in the modeling layer.

While creating content—or simply exploring the data—business users can do on-the-fly aggregations and transformations, intelligent joins, and drill down to lower-level detail such as an individual record. This significantly speeds up development for implementing new requirements and for end users to create their own reports and visualizations.

Last, access and permissions can be handled based on the authenticated user, so even just one dashboard can work for a whole company. No more subsets of data needed per team.

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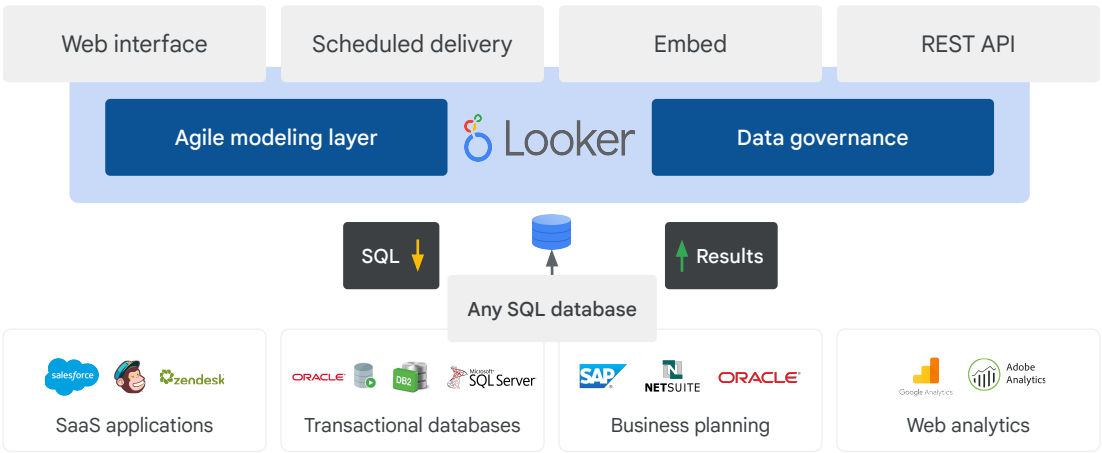
Architecture



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In summary, Looker's unique architecture provides a rich development framework that is built to support enterprise-grade workflows and help your users and tools independently access the most accurate and up-to-date version of your organization's data.

With this unified view into the business, you can choose or design data experiences that ensure both people and systems get data they need, how and when they need it.

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In this section, we'll discuss the paradigm shift in Looker from hardcoding SQL toward the use of LookML, which allows developers to design self-serve data experiences for business users and data analysts.

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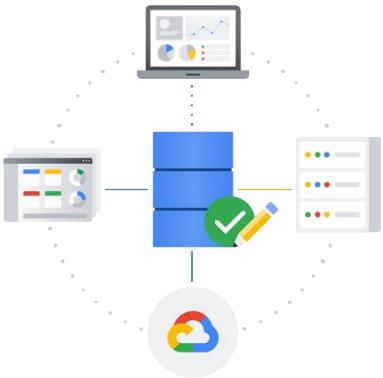
Paradigm shift: Modeling language versus hardcoded SQL



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Looker has two broad categories of users: business users, who are “frontend” users seeking to create and consume content, and developers, who are “backend” users that write LookML code to curate the data provided to frontend users.

LookML provides the backend users, referred to as LookML developers, with the ability to define custom data attributes, aggregates, and calculations as well as model relationships in the source database, without impacting the underlying data.

With LookML, Looker helps to address a few key issues that are common to organizations aiming to leverage data in their day-to-day operations.

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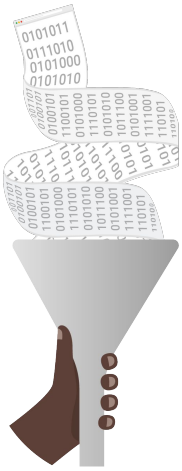
How does LookML drive business outcomes?

Problem 1: Data bottleneck



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The first of the two common operational problems within organizations is the *data bottleneck*. This is where everyone needs access to data, but only a handful have it—typically technical people like data engineers or database administrators.

Business users need to create tickets for reports and often wait days or weeks for them to be fulfilled. If the data engineer misconstrues or mishandles any part of the ticket, the ticket would need to go back to them to correct the report.

The wait time can be frustrating for the business users, and the work can be tedious for the data engineers.

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How does LookML drive business outcomes?

Problem 2: Data chaos

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The second common operational problem is *data chaos*. This is where employees have access to the data, but they are all applying their own formulas and logic.

For example, when Sales, Marketing, and Finance all get into a room, they might not see eye-to-eye on how much revenue the organization made last quarter or how many new customers signed on last month because they were all using different criteria. This can create a culture in which people don’t trust the organization’s data.

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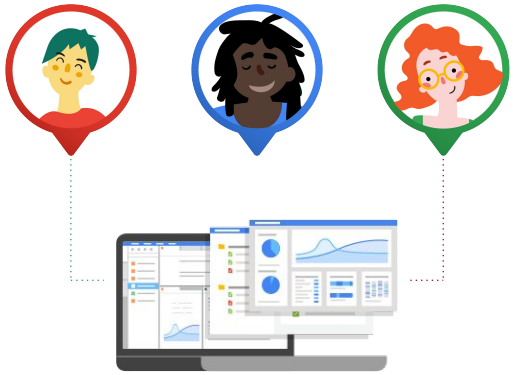
How does LookML drive business outcomes?

Solution: Self-service data + governance



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Looker solves both the problems of the data bottleneck and data chaos through a combination of self-service data and governance.

Using LookML, developers can design and curate Explores, which are report-builder interfaces that offer all of the appropriate fields and logic for analysis and serve as a single source of truth. Business users are then able to easily access the data via the Explores and create their own content such as reports and visualizations.

This enables business users to dive into their data and extract meaningful insights to make data-driven decisions. As for that “handful of technical people” who needed to spend a lot of time on reporting requests and tickets, now they can dedicate their time to other meaningful work that propels the business forward.

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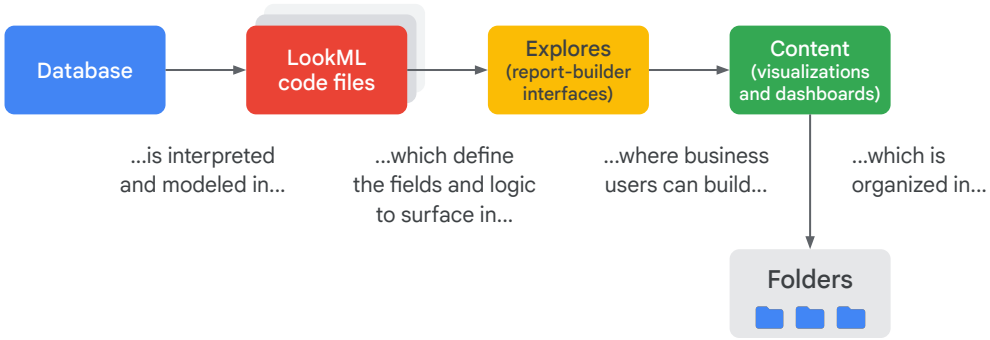
Looker workflow



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This high-level workflow in Looker summarizes how LookML provides self-service data and governance capabilities.

First, the backend database is interpreted and modeled in LookML code files written by LookML developers. Specifically, the LookML code files define the data fields and logic to surface in Explores.

Then, business users build content (which, in Looker, refers to Looks and dashboards), using the Explore data to which they have been granted access by Looker admins.

Last, this content is then organized in folders, to which access can also be controlled as needed.

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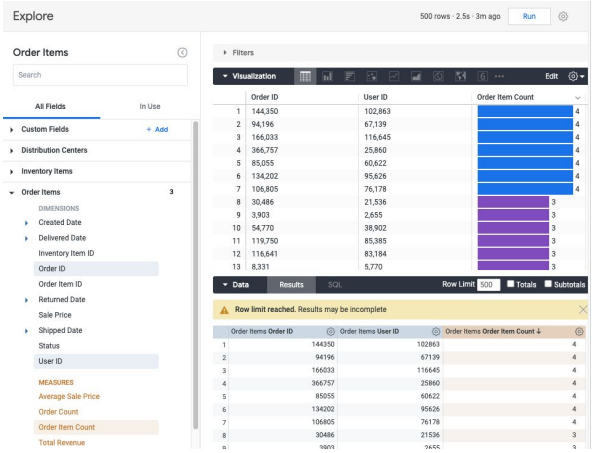
The Explore is where visualizations and dashboards get created

- Report-builder interfaces
- Launch points for visualizations and dashboards
- Convenient, flexible portals to ask questions of the data



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A useful way to think about LookML is that it is an abstraction layer for SQL, which allows developers to tell Looker what data is available in the connected database and how it should interpret that data.

There are many LookML objects such as dimensions, measures, views, models, and more, but for the business user, the end goal is the Explore.

An Explore represents a lot of things:

- It's a report-builder interface, where business users can pick and choose the fields they need for their report.
- It's also a launch point for visualizations and dashboards, meaning every single report, Look, or dashboard tile must come from an Explore.
- And, it's a convenient, flexible portal to ask questions of the data, so business users can look up or analyze something quickly without necessarily making a Look or dashboard out of it.

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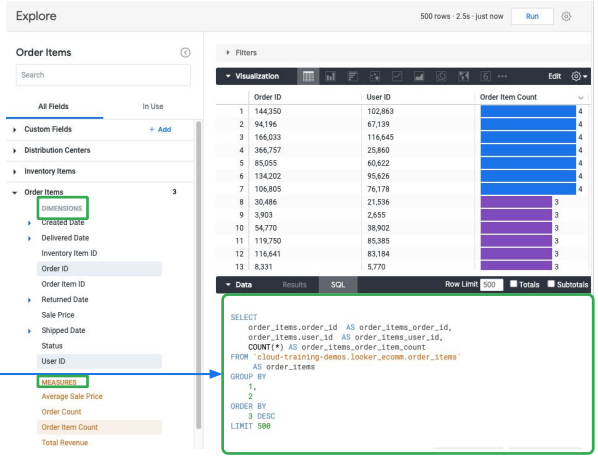
The Explore is where visualizations and dashboards get created

- Single source of truth for field logic and metric calculations
- Auto-generation of SQL queries



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An Explore is also a single source of truth for field logic and metric calculations. Specifically, Explores contain dimensions, which are attributes of data such as a product name or price, and measures, which are aggregates of dimensions, such as the counts, sums, and averages, like the total revenue from sales of a product.

LookML developers define the correct business logic in the agile modeling layer to make the data as flexible as possible and reduce discrepancies, risk, and manual re-work for all the different teams.

Finally, what an Explore does on the backend, what makes all the visualizations and dashboards work, is that it automatically generates SQL queries to send to the connected database. So, with Explores, business users are able to independently explore, analyze, and visualize data without writing any SQL.

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Building SQL from Explores

Typical SQL statement:

```
SELECT column_a, column_b, …, column_n
FROM   table_a as table_a
JOIN   table_b as table_b on table_a.field_1 = table_b.field_1
JOIN   table_c as table_c on table_a.field_2 = table_c.field_2
WHERE  some condition equals some value
```

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To understand how Looker auto-generates SQL queries, let's think about how a typical SQL query works.

First, you SELECT one or more columns FROM a table, and maybe that table needs to be JOINed to other tables. Of course, when you're joining tables, that needs to happen ON one or more key columns.

Then, you might need a WHERE condition to narrow the results.

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Let Looker write the SQL for your analysts

Typical SQL statement:

```
SELECT column_a, column_b, …, column_n
FROM   table_a as table_a
JOIN   table_b as table_b on table_a.field_1 = table_b.field_1
JOIN   table_c as table_c on table_a.field_2 = table_c.field_2
WHERE  some condition equals some value
```

What the Looker model determines:

```
SELECT user-chosen dimensions and measures
FROM   clause plus any JOINs = Explore
WHERE  clause = user-added filters
```

Google Cloud

LookML lets developers define how these parts of SQL statements get generated.

Whatever dimensions and measures the business user chooses are SELECTed as column_a, column_b, and so on.

The structure of the Explore itself determines where the data are selected FROM, which tables might need to be JOINed in, and which columns should be used as the join keys. As such, you can think of an Explore as a set of tables with predefined join logic.

If the business user adds dimension filters, such as “show me only accounts that are active” or “I only want to see support tickets from the year 2020,” these populate the WHERE clause.

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Instead of writing the content of a query, model the structure of the data in Explores

```
SELECT
   flights.destination  AS "destination",
   carriers.name  AS "carriers.name",
   aircraft.name  AS "aircraft.name",
   COUNT(*) AS "flights.1_count"

FROM flights  AS flights

LEFT JOIN public.carriers  AS carriers
      ON flights.carrier = carriers.code

LEFT JOIN public.aircraft  AS aircraft
      ON flights.tail_num = aircraft.tail_num

WHERE (flights.cancelled = 'N')
GROUP BY 1,2,3
```

```
explore: flights {

  join: carriers {
    sql_on: ${flights.carrier} =
${carriers.code} ;;
    relationship: many_to_one
  }

  join: aircraft {
    sql_on: ${flights.tail_num} =
${aircraft.tail_num};;
    relationship: many_to_one
  }
}
```

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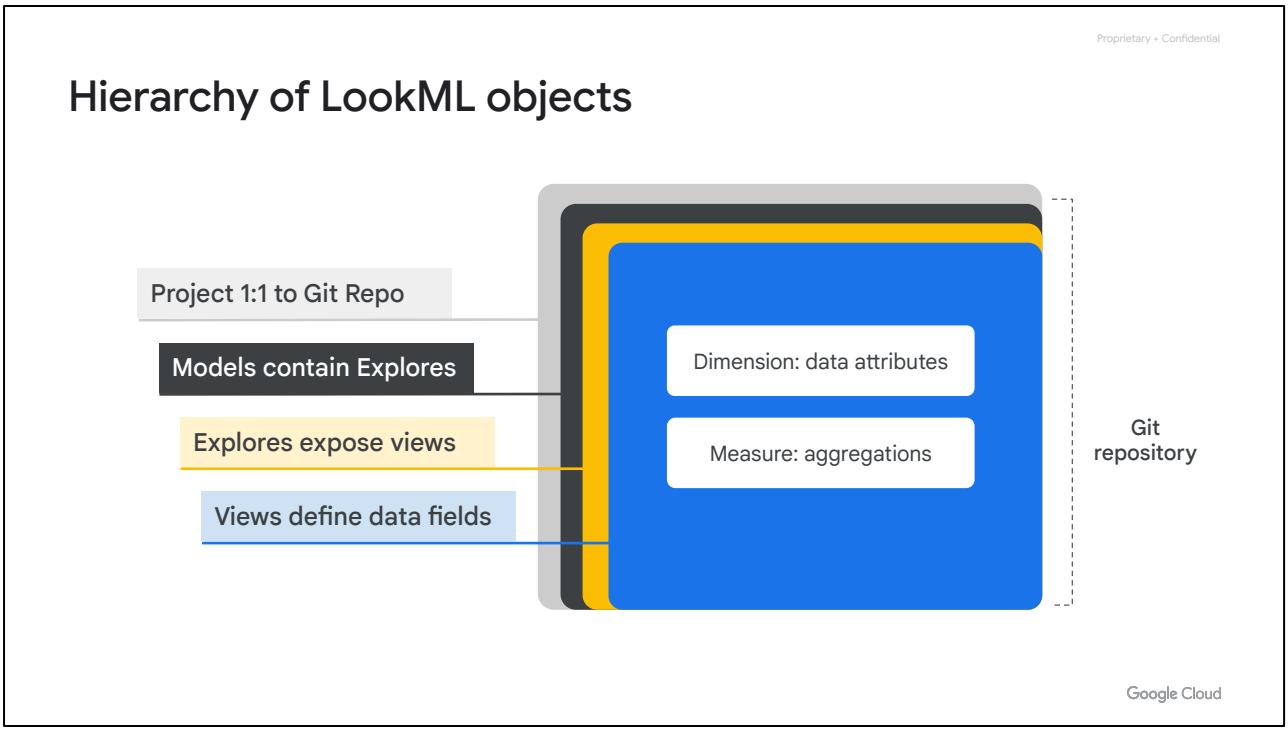
With LookML, gone are the days of having to write long SQL queries (like the one shown on the left) by hand, with multiple selects and joins to get the data that you want.

Imagine what you would have to do if you had built a report with this long SQL query, and a business user said they needed the report rolled up at the monthly or weekly level.

You'd have to add another item to SELECT, maybe look up the syntax to truncate the date to the month or week, maybe also look up the syntax to format the truncated date, then maybe ask the business user if they'd also like a date filter, and don't forget to add that new column to the GROUP BY!

Meanwhile, with the LookML Explore (like the one shown on the right), Looker knows all about the flights, carriers, and aircraft tables. Through the Explore defined by the LookML developer, Looker knows how the tables relate to each other, and what columns and formulas are included in each. Plus, all the date aggregations are already built in.

So as business users select the dimensions and measures that they need for their reports and visualizations, Looker automatically constructs the SQL queries needed and sends them to the underlying database to retrieve the results. No need for business users to write SQL!



While the end goal of most LookML workflows is an Explore, it is useful to know the other components of the LookML hierarchy to understand how Explores are defined.

The highest level of the LookML hierarchy is a project, which is a library of code that models a data source, or SQL database. For effective version control, a project should map 1:1 to a Git repository.

Projects contain specific files that help to define the data in the Explores:

- First, model files, which define the Explores that should be packaged together, how those Explores work, and which views are joined in which Explores.
- Second, view files, which describe database tables or logical representations of them.

The dimensions and measures used for analysis are defined within view files. Recall that In Looker, dimensions are attributes of your data such as a name or location, while measures are aggregates of dimensions such as a count or sum.

These dimensions and measures are then exposed to frontend users through the Explore. These frontend users, often business users or data analysts, can then use the Explore to ~~explore and~~ analyze data as well as create new visualizations and reports.

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What else can we do with LookML?

- Apply conditional *formatting* of colors, styles, etc.
- Link to other areas within or outside of Looker
- Modularize code for easy reuse and expansion
- Define drill-downs and HTML logic
- Restructure data: pre-aggregate values, apply window functions, etc.
- Apply smart caching to conserve database resources/bandwidth
- Configure row- and column-level security on the underlying data
- Create dynamic logic for aggregations, filters, joins, etc.

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Although the Explore is the primary end product that LookML developers create for business users, to say LookML is only about building Explores would be a vast oversimplification. You can do lots of cool things in LookML!

Here are just some ideas.

- For example, you can apply conditional formatting of colors, styles, and more.
- Link to other areas within or outside of Looker.
- Modularize code for easy reuse and expansion.
- You can also define drill-downs—or how people can get more granular into their data—and HTML logic to customize the way results are displayed.
- Restructure data by pre-aggregating values, applying window functions, and more.
- Apply smart caching to conserve database resources and bandwidth.
- Configure row- and column-level security on the underlying data.
- And create dynamic logic for aggregations, filters, joins, and more.

With all of these capabilities, LookML makes it easy for developers to curate custom data experiences for business users at your organization.

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Google Cloud

In this section, we'll review the core analytical concepts that will enable you to start analyzing and visualizing data in Looker.

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The four core analytics concepts

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As you start to explore, analyze, and visualize data in Looker, there are four core analytics concepts that you need to know, whether you are a front-end user or a LookML developer. If you have used another BI tool, these may already be familiar to you.

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The four core analytics concepts

Dimensions

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First, dimensions are qualities or attributes of your data. For example, if you are looking to buy a new gadget online, you might want to know its brand name, color, and retail price. Each of these would be a dimension of the gadget.

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Example dataset

Fruit Type	Color	Is Round	Price per Pound	Weight	Price
apple	red	yes	$1.25	0.6	$0.75
apple	red	yes	$1.25	0.55	$0.69
apple	green	yes	$1.15	0.4	$0.46
banana	yellow	no	$0.75	0.7	$0.53
banana	yellow	no	$0.75	0.75	$0.56
orange	orange	yes	$1.45	0.5	$0.73
orange	orange	yes	$1.45	0.45	$0.65
lemon	yellow	no	$2.25	0.2	$0.45
lemon	yellow	no	$2.25	0.22	$0.50
lemon	yellow	no	$2.25	0.19	$0.43
lime	green	no	$2.40	0.18	$0.43
lime	green	no	$2.40	0.17	$0.41

Google Cloud

In this example, the table contains 12 rows that describes the fruit basket on the left. Each piece of fruit is represented as a row in the table, with each column describing a different attribute or aspect of the fruit.

Each column in the database table is a Looker dimension. Specifically, each fruit has a type, has a color, is either round or not round, has a price per pound, has a weight, and has a unit price.

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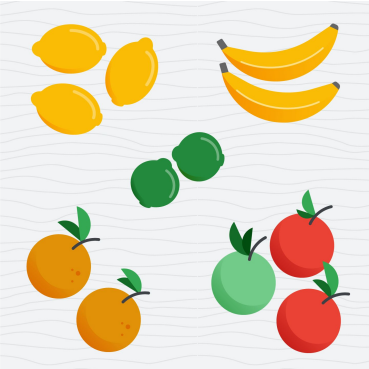
Dimensions: The columns of the table

What types of fruit are in the basket?



Google Cloud

Imagine that you want to know, “What types of fruit are in the basket?” What is the dimension you need here?

One that identifies **Fruit Type**.

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Dimensions: The columns of the table

Choosing the Fruit Type dimension will return the unique fruit names from the data.

Looker

Fruit Basket Fruit Type

1	orange
2	apple
3	banana
4	lemon
5	lime

5

Google Cloud

When you select a dimension in Looker, only the unique possible values are displayed.

As such, the **Fruit Type** dimension returns 5 rows for the 12 fruits: one each for oranges, apples, bananas, lemons, and limes.

Even though in the original fruit basket that there were 3 lemons, 2 bananas, and so on, you don't get multiple rows for them; in these results for the **Fruit Type** dimension, you have only one row for each.

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The four core analytics concepts

- Dimensions
- Measures

Google Cloud

The second core analytics concept in Looker is a measure. Measures are aggregate calculations applied across one or many rows of data.

For example, what is the total cost for 2 or 3 electronic gadgets? How much is the cheapest one? How many do we have in our cart?

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Example dataset

Fruit Type	Color	Is Round	Price per Pound	Weight	Price
apple	red	yes	$1.25	0.6	$0.75
apple	red	yes	$1.25	0.55	$0.69
apple	green	yes	$1.15	0.4	$0.46
banana	yellow	no	$0.75	0.7	$0.53
banana	yellow	no	$0.75	0.75	$0.56
orange	orange	yes	$1.45	0.5	$0.73
orange	orange	yes	$1.45	0.45	$0.65
lemon	yellow	no	$2.25	0.2	$0.45
lemon	yellow	no	$2.25	0.22	$0.50
lemon	yellow	no	$2.25	0.19	$0.43
lime	green	no	$2.40	0.18	$0.43
lime	green	no	$2.40	0.17	$0.41

Google Cloud

Here’s the fruit basket dataset again.

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Measures: The math used to explore the data

How many pieces of fruit are in the basket?

Looker

Google Cloud

Maybe this time, you want to know “How many pieces of fruit are in the basket?” Any time you have a question like “how many,” you probably want to look for a measure with the word **Count** or **Number** in it.

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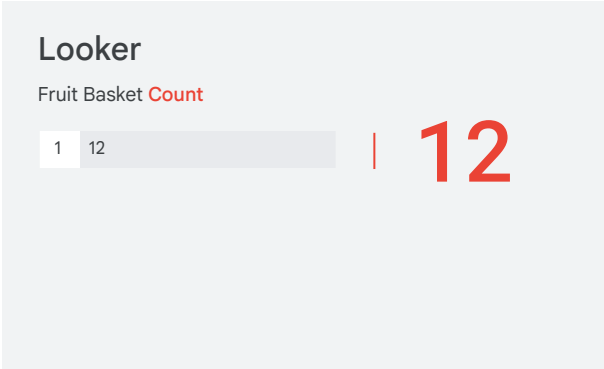
Measures: The math used to explore the data

Choosing the Count measure will tell us how many pieces of fruit we have.



Google Cloud

In this case, you need the **Count** measure, which tells you that there are 12 pieces of fruit.

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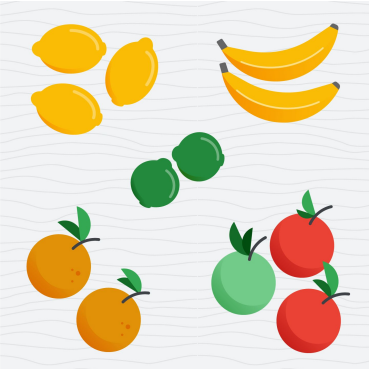
Dimensions and measures

How many pieces of each fruit type are in the basket?



Looker

Google Cloud

Now, you may also want to know “How many pieces of *each* fruit type are in the basket?” How do you break down this question?

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Dimensions and measures

Choosing the Fruit Type dimension and the Count measure will tell us how many of each type of fruit we have.

Looker

Fruit Basket Fruit Type | Fruit Basket Count

1	apple	3
2	lemon	3
3	lime	2
4	banana	2
5	orange	2

Google Cloud

"How many pieces" suggests that you need the **Count** measure again, while "of each fruit type" means you need the **Fruit Type** dimension.

Putting the Fruit Type dimension and the Count measure together, you get the five rows because Looker grabs only the unique possible values of the dimension.

You also see the overall count of 12, split out across those five rows: 3 apples, 3 lemons, and so on.

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The four core analytics concepts

- Dimensions
- Measures
- Filters

Google Cloud

The next core analytics concept in Looker is a filter. Filters are ways to reduce or narrow down the results returned. They don't delete anything from the database; they're only applied to the data that Looker displays on your screen.

For example, when shopping online, you might only want to see the gadgets that are blue (which would be a dimension filter), or that have an average rating of 4 out of 5 (which would be a measure filter).

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Example dataset

Fruit Type	Color	Is Round	Price per Pound	Weight	Price
apple	red	yes	$1.25	0.6	$0.75
apple	red	yes	$1.25	0.55	$0.69
apple	green	yes	$1.15	0.4	$0.46
banana	yellow	no	$0.75	0.7	$0.53
banana	yellow	no	$0.75	0.75	$0.56
orange	orange	yes	$1.45	0.5	$0.73
orange	orange	yes	$1.45	0.45	$0.65
lemon	yellow	no	$2.25	0.2	$0.45
lemon	yellow	no	$2.25	0.22	$0.50
lemon	yellow	no	$2.25	0.19	$0.43
lime	green	no	$2.40	0.18	$0.43
lime	green	no	$2.40	0.17	$0.41

So how might you filter the fruit basket?

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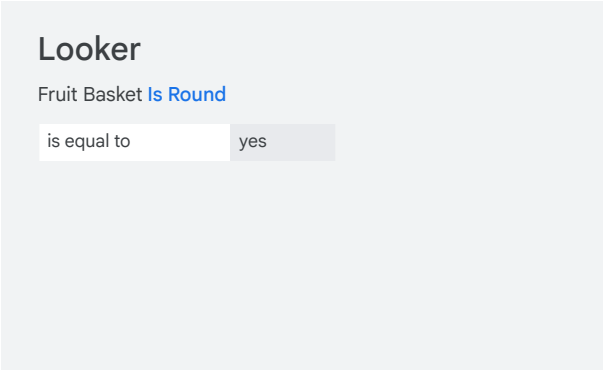
Dimensions as filters

Filtering a dimension removes rows from the raw data.



Google Cloud

Maybe you want to know, “How many pieces of each fruit type are there if I only include round fruits?”

You could apply a filter on the **Is Round** dimension, and set it equal to “yes”, and then, select the Count measure.

This is called applying a *dimension filter*, and what your database does behind the scenes is to figure out which rows of data need to be included or excluded right from the start.

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Dimensions as filters

The filter on the dimension is applied first, and then the data is aggregated.

Looker

Fruit Basket Is Round

is equal to	yes

Fruit Basket Fruit Type | Fruit Basket Count

1	apple	3
5	orange	2

Google Cloud

Behind the scenes, Looker first skips over or ignores the fruits that aren't round, since they don't meet the criterion of our dimension filter.

Then it takes what's left—apples and oranges—and performs the aggregation, so you get 3 apples and 2 oranges.

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Filter measures

Which types of fruit have exactly two pieces in the basket?

Looker

Google Cloud

Now, what if you want to know “Which types of fruit have exactly two pieces in the basket?”

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Filter measures

Filtering a measure removes rows from the result set, not from the raw data.

Looker

Fruit Basket Count

is equal to	2

Fruit Basket Fruit Type | Fruit Basket Count

1	apple	3
2	banana	2
3	lemon	3
4	lime	2
5	orange	2

Google Cloud

You would need to apply a filter on the **Count** measure, setting it to equal 2.

The key difference when filtering dimensions versus filtering measures lies in when the database decides to discard results.

With dimension filters, Looker applies the filter at the very beginning. With measure filters, Looker first gets all the possible rows and calculates the aggregation for each...

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Filter measures

After filtering a measure, we are left with all rows in the result set that fit the filter logic.

Looker

Fruit Basket Count

is equal to	2

Fruit Basket Fruit Type | Fruit Basket Count

1	banana	2
2	lime	2
3	orange	2

Google Cloud

… and then discards the inapplicable results at the *end*.

In this case, the final results only show banana, lime, and orange, which are the fruit types with a count equal to 2.

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The four core analytics concepts

- Dimensions
- Measures
- Filters
- Pivots

Google Cloud

Last, pivots turn a dimension into several columns, like pivot tables in spreadsheet software. Your data would basically become a matrix, so you can analyze groupings of data.

In the shopping scenario, you might want to see a list of offering levels like “basic,” “deluxe,” and “premier,” and then *pivot on* brand name. This might let you see for each level and brand name, how many gadgets are out there, their average rating, and so on.

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Display results with pivots

Pivoting enables dimensions to be displayed horizontally. Pivoting on Is Round creates new column headers based on the unique values in Is Round.

Looker

Fruit Basket Color | Fruit Basket Is Round | Fruit Basket Count

yellow	no	5
orange	yes	2
red	yes	2
green	no	2
green	yes	1

Google Cloud

Now, imagine that you want to know, “How many pieces of fruit are there for each color, and is the fruit round or not?”

This display is adequate; it’s easy enough to see that there are 5 fruits that are yellow and not round, 2 that are orange and round, etc.

But it’s not easy to compare all the colors against each other, or just the ones that are round against each other.

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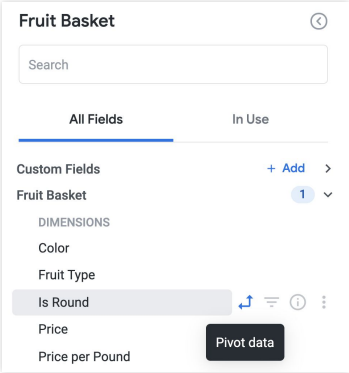
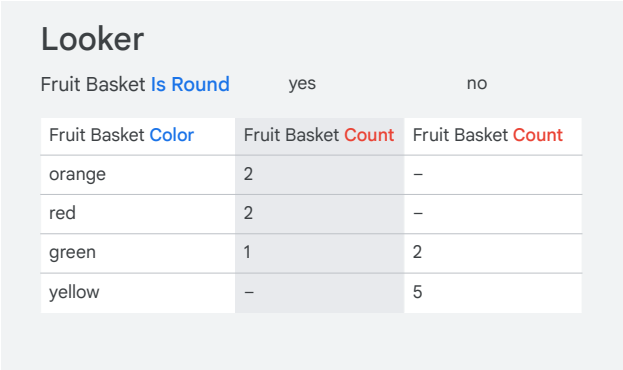
Display results with pivots

Pivoting enables dimensions to be displayed horizontally. Pivoting on Is Round creates new column headers based on the unique values in Is Round.





Looker

Fruit Basket Is Round	yes	no
Fruit Basket Color	Fruit Basket Count	Fruit Basket Count
orange	2	–
red	2	–
green	1	2
yellow	–	5

Google Cloud

Applying a pivot to the **Is Round** dimension transforms the display of the data in a way that is more meaningful.

When you pivot on a dimension, each unique possible value of that dimension becomes its own column header. Any measures are then repeated under each column header.

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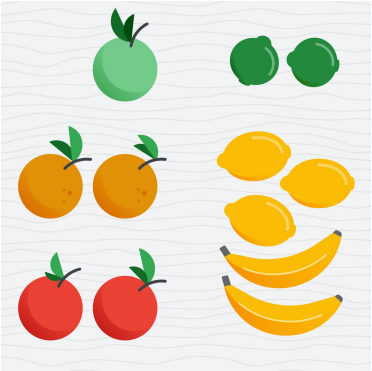
Display results with pivots

The Is Round categories yes and no become our new column headers. All of the round fruit is in the first column.



Looker

Fruit Basket Is Round	yes	no
Fruit Basket Color	Fruit Basket Count	Fruit Basket Count
orange	2	–
red	2	–
green	1	2
yellow	–	5

Google Cloud

Now, you have one column for **Is Round** being “yes” and another for “no”.

It becomes much easier to compare just the counts of round fruit across colors, and the counts of the fruit that are not round.

It also shows you gaps in the data, where there aren’t *any* round, or not-round, fruit for a particular color.

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The four core analytics concepts

- Dimensions
- Measures
- Filters
- Pivots

Google Cloud

Now, after this brief introduction to the four core analytics concepts, you are ready to start analyzing and visualizing data in Looker.

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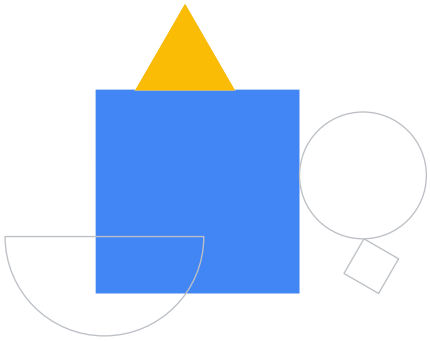
Lab Intro

Getting Started with Analyzing and Visualizing Data in Looker



Google Cloud

In this lab, you learn how to apply the four core analytics concepts in Looker to analyze and visualize data, using Explores that have been created by LookML developers.

The lab walks you through key tasks that are frequently completed by data analysts and business users in a Looker instance.
Specifically, the objectives of this lab include:

- Using the Explore interface to access data curated by LookML developers.
- Working with dimensions, measures, filters, and pivots to query and select data.
- Selecting the appropriate visualization type to best display your data.
- Saving your visualizations to dashboards.